

Explainer: Problem 4 (Rudin 7.1) — Uniform Convergence $\Rightarrow$ Uniform Boundedness

What's being claimed?

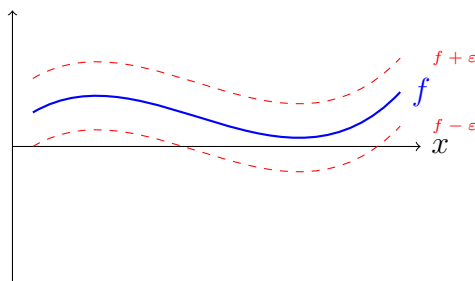
If $\{f_n\}$ is a sequence of bounded functions (each $|f_n(x)| \leq M_n$ for some constant M_n) and $f_n \rightarrow f$ **uniformly**, then there is a *single* constant M with $|f_n(x)| \leq M$ for *all* n and *all* x .

Two kinds of boundedness

- Each f_n bounded: for each n , some M_n works. But M_n could blow up in n .
- **Uniformly bounded:** one single M works for the whole sequence.

Why uniform convergence helps

Uniform convergence means: eventually, *all* f_n are within ε of f *everywhere*. Once you're close to f , you inherit f 's bound. The only worry is the finitely many terms *before* that point — but finitely many bounded functions are jointly bounded (take the max).



For $n \geq N$, every f_n lies in the red strip.

Strategy

1. Use uniform convergence: pick N so that $|f_n(x) - f(x)| < \varepsilon$ (any ε) for all $n \geq N$ and all x .
2. f is bounded (as a uniform limit of bounded functions — prove this first!).

3. For $n \geq N$: $|f_n(x)| \leq |f(x)| + 1 \leq \|f\|_\infty + 1$.
4. For $n < N$: just finitely many; each has its own bound M_n .
5. Take $M = \max(M_1, \dots, M_{N-1}, \|f\|_\infty + 1)$.

A subtle step: the limit is bounded

You'll need $|f(x)| \leq \|f_N\|_\infty + \varepsilon$ for all x , which follows from $|f(x) - f_N(x)| < \varepsilon$ and $|f_N(x)| \leq \|f_N\|_\infty$.